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PGA-UNet: A Lightweight Prompt-Guided Attention U-Net for Interactive Bone X-Ray Lesion Segmentation

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ABSTRACT Bone X-ray lesion segmentation is a challenging task because lesions are often small, low-contrast, and visually entangled with the surrounding anatomy. In this setting, the main challenge typically lies in localizing the suspicious region rather than refining the lesion's boundary. A practical strategy is to narrow the search space before segmentation itself is performed. We realize this idea through a direct interactive approach: a clinician supplies a coarse bounding box, which the model then uses as a spatial guidance signal.
We propose PGA-UNet, a lightweight prompt-guided CNN that converts the bounding-box prompt into a Gaussian-smoothed plateau heatmap, modulates encoder features according to the prompt through a Prompt Spatial Gate (PSG), and sustains prompt influence in the decoder through a Conditional Attention Decoder (CAD). The goal is not only to use the prompt to indicate where segmentation should occur, but also to keep small-lesion-relevant features active throughout the network.
Attention U-Net without a prompt serves as an automatic reference, and SAM-Med2D as a prompt-matched reference. On BTXRD and FracAtlas, evaluated at 128, 256, and 512 resolutions under controlled covering and off-center boxes, prompt-free segmentation is difficult, with Attention U-Net reaching only 0.52 and 0.34 Dice. Given the same box, PGA-UNet is consistently ahead of two prompt-matched conventional baselines and of fine-tuned SAM-Med2D, with the margin largest on the harder FracAtlas data and on the smallest lesions, where fine-tuned SAM-Med2D drops to 0.26 to 0.37 Dice while PGA-UNet holds 0.65 to 0.72. Repeated random splits show the trend is stable, and PGA-UNet uses about 92 times fewer parameters than SAM-Med2D. A training-free score built from mask decisiveness and stability under prompt jitter gives a usable no-ground-truth ranking of prediction quality, with a rank correlation to the true Dice of roughly 0.5 to 0.7.
Because all boxes are simulated around annotated lesions, the evidence concerns prompted segmentation after controlled localization, not unconstrained detection, partial or off-target boxes, calibrated confidence, clinical usability, patient-level generalization, or cross-dataset transfer. A repeated-seed ablation of the individual components is still in progress.

INDEX TERMS bone X-ray, interactive segmentation, medical image segmentation, prompt-guided learning, Attention U-Net

I. INTRODUCTION

BONE X-ray imaging is widely used in screening, diagnostic support, and follow-up because of its speed, low cost, and broad accessibility. Lesion segmentation in radiographs, however, remains difficult when the lesion is very small, low-contrast, or partially obscured by surrounding anatomy. In these cases, one of the main challenges lies in localization: before any boundary can be refined, the model must first identify the suspicious region within the entire image.

This is why small-lesion segmentation is rarely just a boundary-delineation problem. When a lesion occupies only a tiny fraction of the image, searching the entire radiograph is both inefficient and error-prone. A natural response is to narrow the search space first, then segment within the reduced region. Fully automatic methods attempt to learn this process internally. Although U-Net and Attention U-Net [3], [4] remain representative models of this approach, they still require the network to localize and segment the lesion entirely on its own, without any external assistance.

An alternative is to draw region-of-interest guidance from an external source. In clinical practice, a clinician can easily draw a coarse bounding box around a suspicious area, something a fully automatic model can rarely detect reliably for very small lesions. In this sense, a bounding-box prompt is more than an interactive convenience; it is a practical means of injecting spatial knowledge into the pipeline before detailed segmentation begins. Prompt-based interactive segmentation methods have grown out of this idea, from clicks and scribbles [5], [6] to recent promptable foundation models such as SAM and SAM-Med2D [1], [2].

For the present task, however, one issue remains unresolved. Once a prompt has identified the region of interest, should it serve merely as a positional constraint for mask generation, or should it continue to shape feature extraction so that faint lesion signals survive downsampling? This distinction matters most for very small lesions, where a small spatial deviation can substantially affect the segmentation outcome.

We address this question with PGA-UNet, a lightweight prompt-guided CNN in which the box prompt is treated as a structured spatial prior rather than an auxiliary input channel. The prompt is encoded as a plateau-style Gaussian heatmap, which is then modulated into encoder features through a Prompt Spatial Gate (PSG) and reused in the decoder through a Conditional Attention Decoder (CAD). This design is intended to keep prompt-guided lesion features active throughout the network, rather than merely confining segmentation to a predefined box region.

This perspective also shapes how the experiments are interpreted. Because Attention U-Net receives no spatial guidance from a clinician, its role is only to show how difficult the problem becomes when localization must be fully automatic. SAM-Med2D, in contrast, is chosen as the direct comparison because it receives the same bounding-box prompt input. The central question here is therefore not whether prompts are useful in general, but whether a lightweight architecture can exploit a coarse clinician prompt more effectively by preserving weak lesion signals, rather than treating the prompt merely as a spatial constraint.

The main contributions of this article are as follows.

- We propose PGA-UNet, a lightweight prompt-guided architecture for interactive bone X-ray lesion segmentation.
- We design a prompt-conditioning mechanism that combines a plateau-style Gaussian prompt, an encoder-side Prompt Spatial Gate (PSG), and a decoder-side Conditional Attention Decoder (CAD).
- We formulate an offline training and online interactive inference framework, with explicit mathematical definitions for prompt encoding, encoder gating, and decoder conditioning.
- We evaluate the method on the BTXRD and FracAtlas datasets under covering and off-center prompt conditions, comparing against an automatic Attention U-Net, two prompt-matched Attention U-Net baselines that receive the same box, and SAM-Med2D, using Dice and IoU for region overlap, a center-based localization score (CBL), and HD95 for boundary distance, together with Monte Carlo cross-validation and computational efficiency measurements.
- We show, through a small-lesion analysis and an Attention U-Net-defined difficulty split, that the benefit of prompt-conditioned processing is concentrated where automatic localization fails and where lesions are smallest.
- We describe a training-free self-assessment score, built from mask decisiveness and stability under prompt jitter, that gives a usable no-ground-truth ranking of prediction quality and can shortlist candidate regions for clinician review.

II. RELATED WORK

Small-lesion segmentation in medical imaging has long been constrained by a simple practical fact: before a model can segment a tiny target well, it must first narrow down where to look. In the fully automatic route, this search-space tightening is learned implicitly inside the network. U-Net established the canonical multi-scale encoder-decoder template for biomedical segmentation [3], and Attention U-Net later added attention gates on the skip connections so that decoder-side fusion could suppress irrelevant spatial responses more selectively [4]. Self-configuring pipelines such as nnU-Net push the same route further by adapting preprocessing, network shape, and training schedule to each dataset automatically [10]. These models raised automatic segmentation quality considerably, but they still assume that the network itself localizes and delineates the lesion from the raw image alone, with no external hint about where the target lies.

Interactive medical segmentation developed in parallel, from the observation that a small amount of user guidance can simplify the problem substantially [5], [6]. Early methods used clicks, scribbles, and geodesic cues; later work added boxes as a fast coarse prompt. More recent interactive models such as ScribblePrompt handle several prompt types within a single framework and across many biomedical datasets [11]. Conceptually, all of these methods move part of the localization burden from the model to the user. For tiny radiographic lesions that shift is especially valuable: a clinician can usually mark a suspicious region coarsely even when a fully automatic system would struggle to find it reliably in the whole image.

Promptable foundation models pushed this further. SAM made prompt-driven segmentation general [1], SAM-Med2D and MedSAM carried it into medicine [2], [9], and adapter variants such as EMedSAM and Med-SA cut the cost of specializing those models to medical data [12], [13]. They are flexible, but they consume the prompt mainly at the mask

TABLE 1. Representative prior work and the remaining gap addressed in this article.

Work	Principle	Method	Performance Evidence	Pros	Cons
U-Net [3]	Fully convolutional encoder-decoder	Multi-scale skip-connected segmentation network	Widely used as a canonical biomedical segmentation baseline	Simple, efficient, strong reference model	No prompt guidance; all localization must be automatic
Attention U-Net [4]	Skip-feature filtering by attention gates	U-Net with gated skip connections	Medical segmentation tasks reported with overlap metrics such as Dice and IoU	Stronger automatic baseline than plain U-Net	Attention is still learned without explicit prompt conditioning
nnU-Net [10]	Self-configuring automatic pipeline	Dataset-adaptive preprocessing, architecture, and training	Strong automatic results across many medical benchmarks	Robust automatic pipeline with little manual tuning	No prompt mechanism; localization remains fully automatic
DeepIGeoS and related interactive methods [6]	User guidance reduces search space	Geodesic or interaction-aware segmentation	Interactive medical segmentation under user input	Better reflects clinical interaction	Interaction mechanism can be task-specific or costly to maintain
ScribblePrompt [11]	General interactive biomedical segmentation	One model for scribbles, clicks, and boxes across datasets	Multi-dataset interactive evaluation with several prompt types	Flexible, realistic interaction types	General-purpose, not a lightweight radiograph-specific prompt prior
SAM-Med2D / MedSAM [2], [9]	Promptable foundation segmentation	Large pretrained model with prompt-driven mask decoding	Multi-dataset medical evaluation with prompt-based metrics	Flexible and broadly applicable	Heavy; prompt influence is not designed as a lightweight end-to-end radiograph-specific prior
EMedSAM / Med-SA [12], [13]	Adapter tuning of SAM for medical images	Lightweight adapters on a frozen SAM backbone	Medical segmentation with reduced adaptation cost	Cheaper specialization of a foundation model	Still a large SAM backbone; prompt acts mainly at the mask decoder
This work	Prompt as a dense spatial prior throughout the network	Gaussian plateau prompt + PSG + CAD in a lightweight CNN	BTXRD and FracAtlas; Dice, CBL, HD95, efficiency, and small-lesion analysis	Lightweight, prompt-aware from encoder to decoder, evaluated under controlled prompt displacement	Still requires supervised masks and fixed-resolution preprocessing

decoder, they carry a large backbone, and none is tuned to a single radiograph task. That leaves one question open for very small bone lesions: once the prompt has localized the region, should it merely mark where the mask goes, or should it keep shaping feature extraction so a weak lesion signal survives downsampling?

That question is where this work starts. In grayscale bone radiographs, knowing roughly where the lesion is only gets you so far; the harder part is holding on to faint evidence while suppressing clutter after the search has already been narrowed. A lightweight prompt-guided CNN might do better by keeping the prompt live from the first encoder stage to the last decoder stage, rather than using it once and letting it fade. We study that setting directly, with SAM-Med2D as the promptable reference; a wider comparison against newer promptable medical models is left for future work.

III. METHOD

A. OFFLINE AND ONLINE FRAMEWORK

The proposed system has two stages. In the offline stage, the model is trained on radiographs paired with lesion polygons. Each polygon is converted into a training prompt by generating a simulated bounding box and then transforming that box into a Gaussian-smoothed plateau heatmap. The network learns to map the image-prompt pair to the binary lesion mask. In the online stage, a user provides a coarse bounding box around a suspected lesion on a new radiograph. The same prompt-construction procedure is applied, and the trained model predicts the final lesion mask.

What sets this apart is where the prompt lives. It is not a side input consumed at the first layer and then forgotten. It becomes a dense spatial prior that stays active through the encoder and the decoder, which is what should help when the box is coarse, slightly off-center, or wrapped around a tiny lesion.

An interactive demonstration of the online stage is shown in Figure 1: the user draws a box on a radiograph, and the model returns the mask together with the no-ground-truth CAD prompt-use score and the self-assessment score of Section III-F.

B. PROBLEM SETTING

Given a grayscale radiograph $\mathbf{I} \in \mathbb{R}^{H \times W}$ and a user-provided bounding box $\mathbf{B}$, the goal is to predict a binary lesion mask $\hat{\mathbf{M}}$. Let $\mathbf{H}(\mathbf{B})$ be the prompt heatmap derived from the box. PGA-UNet predicts

$$\hat{\mathbf{M}} = \mathbf{1}\left[f_{\mathrm{PGA}}\left(\mathbf{I}, \mathbf{H}(\mathbf{B}); \theta\right) \geq 0.5\right]. \tag{1}$$

C. PROMPT REPRESENTATION

The input prompt is not encoded as a hard binary rectangle. Instead, the box interior is rasterized into a plateau map at the original image resolution and its boundaries are softened with a fixed 31×31 Gaussian kernel, before the heatmap is resized and padded down to the target square resolution together with the image. The kernel size is not rescaled with the target resolution (256×256 or 512×512): applying it once at the original resolution, prior to downsampling, keeps the effective blur consistent relative to whichever square frame the network ultimately sees. The plateau covers exactly the (possibly expanded) box with no additional minimum margin; coverage of the region of interest is guaranteed by the box expansion described below, not by a separate margin step.

During evaluation, prompts are tested in two fixed settings. The covering prompt (`center_zoom`) scales the tight lesion box outward from its own center by a fixed factor of 3.0. The off-center prompt (`center_shift`) applies the same scaling and then displaces the box by a fraction of the tight-box size bounded by `shift_ratio`=0.5, after which the box is expanded back if needed so that it still fully contains the lesion. At test time this displacement is deterministic per lesion, seeded from the sample index, so every model is evaluated on exactly the same off-center boxes; during training the displacement is redrawn randomly at each iteration. A single model is trained per dataset and resolution with `center_mixed`, which selects `center_shift` with probability 0.8 and `center_zoom` with probability 0.2, and the two conditions are scored separately at test time. The task scope assumes that the user has already identified the suspicious region and provides a box meant to cover that lesion; partial-coverage, negative, and unrelated-region boxes lie outside this controlled prompted-segmentation protocol, as does the separate question of whether a clinician can find the correct region in the first place.

Let (x_1, y_1, x_2, y_2) denote the tight lesion box and let $w = x_2 - x_1$ and $h = y_2 - y_1$. The simulated covering box uses the fixed center scaling described above during training and evaluation. The off-center box uses the same scaled box and applies the deterministic test-time displacement rule implemented in `dataset.py`, while preserving lesion coverage. The scale factor, the displacement ratio, and the Gaussian kernel size were chosen from preliminary experiments on the training and validation data and then frozen; the test split was

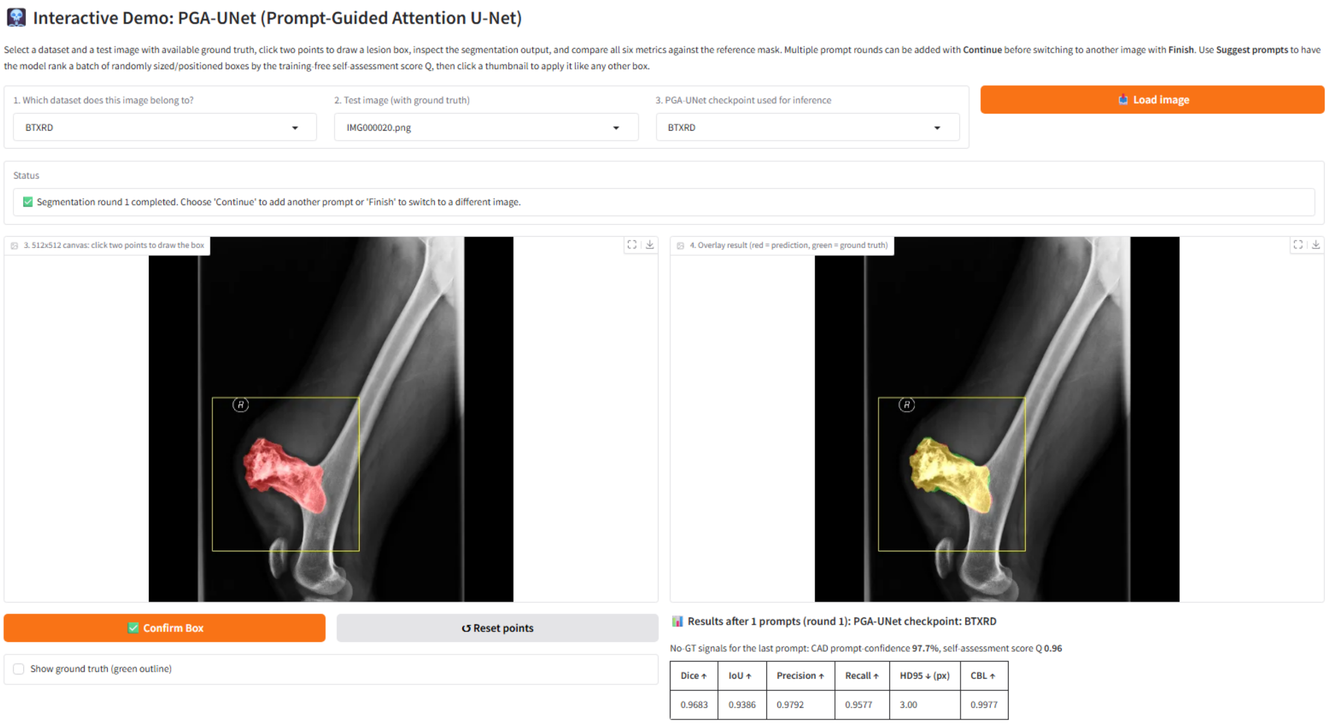

FIGURE 1. **Interactive demonstration of the online stage on a BTXRD case. The clinician draws a box on the left canvas; the right panel overlays the predicted mask, and the panel below reports the reference metrics together with the no-ground-truth CAD prompt-use score and the self-assessment score *Q*. A FracAtlas case is shown in the same layout in the supplementary material.**

not used for any protocol or hyperparameter choice. They are reported as fixed protocol values, not as globally optimal settings. This simulation is not a substitute for radiologist-drawn prompts, but it provides a controlled way to measure sensitivity to the defined prompt displacement.

Algorithmically, prompt construction follows three steps: 1) derive a lesion box from the polygon annotation or user input and expand it as described above, 2) rasterize the expanded box into a plateau mask at the original image resolution, and 3) smooth the plateau boundary with the fixed Gaussian kernel before resizing to the target resolution. This sequence avoids the hard-edge dependence of a binary box while preserving a clear spatial prior.

D. PROMPT SPATIAL GATE

Let $\mathbf{x}^l$ denote the encoder feature map at level l. PSG modulates it with a prompt-derived attention map $\mathbf{A}^l_{\text{PSG}}$:

$$\widetilde{\mathbf{x}}^l = \mathbf{x}^l \odot \left(1 + \alpha^l_{\text{PSG}} \mathbf{A}^l_{\text{PSG}}\right), \tag{2}$$

where $\odot$ is element-wise multiplication and α^l_{PSG} is learnable. In the implementation, α^l_{PSG} is initialized to 0.1 and clamped to $[0, 1]$. The residual form keeps the original feature stream available while emphasizing prompt-relevant regions.

E. CONDITIONAL ATTENTION DECODER

CAD extends decoder attention by conditioning the skip gate on prompt-encoded features. Let $\mathbf{g}^l$ be the decoder gating feature and $\mathbf{p}^l_{\text{enc}}$ the prompt encoding at the same scale. The conditioned gate is

$$\mathbf{g}'^{\,l} = \mathbf{g}^l + c^l \alpha^l_{\text{CAD}} w_l \mathbf{p}^l_{\text{enc}}, \tag{3}$$

where c^l is a decoder-derived prompt-use scalar, α^l_{CAD} is learnable, and w_l is a fixed depth coefficient. This design is intended to keep the decoder conditioned on the prompt when the box is not perfectly centered.

The prompt encoder in CAD uses two 3×3 convolutions with instance normalization and ReLU. The prompt-use scalar c^l is produced by global average pooling followed by a 1×1 convolution and a sigmoid nonlinearity. It is computed inside the gate from the prompt encoding and is used only to scale the prompt term at that level; it is not exposed or analyzed as a separate model output in this article. The decoder depth coefficients are fixed to $\{1.0, 0.7, 0.4, 0.2\}$ from deep to shallow layers. With feature scale 4, the channel widths are $\{16, 32, 64, 128, 256\}$ from the shallowest encoder stage to the bottleneck, and the complete model has approximately 2.95 million parameters. These protocol values were selected in preliminary experiments on the training and validation data and held fixed for the main evaluation, with the test split excluded from any tuning; no optimality claim is made. The CAD mixing coefficient is parameterized as a sigmoid-transformed scalar and initialized near 0.3.

Taken together, the method can be summarized as follows: 1) preprocess the image and prompt into a common square resolution, 2) encode prompt-relevant spatial priors

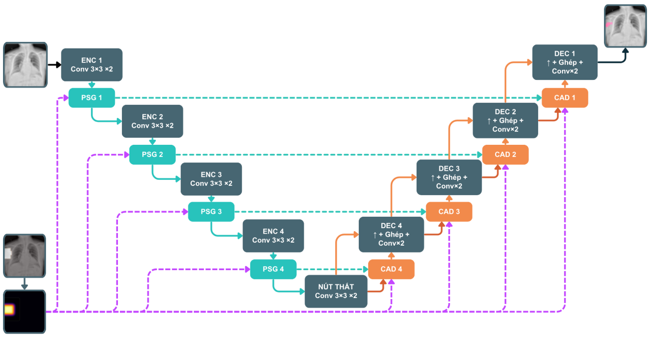

FIGURE 2. Overview of PGA-UNet. Prompt information derived from the bounding box is injected into encoder features through PSG and reused in decoder skip attention through CAD.

with PSG in the encoder, 3) propagate prompt-conditioned context through CAD in the decoder, and 4) predict the final binary lesion mask. The scientific meaning of the contribution is that prompt information is both spatially softened and repeatedly reused, rather than being injected once and left to dissipate.

F. TRAINING-FREE SELF-ASSESSMENT SCORE

PGA-UNet also returns a scalar Q that estimates, with no ground truth, how far a given prediction can be trusted. Nothing about Q is learned. It is read off the model's own output at inference as the average of two cues.

The first cue, S_{sharp}, measures how decisive the predicted mask is. Let p be the sigmoid probability map and $\Omega = \{p > 0.5\}$ the predicted foreground. Then

$$S_{\text{sharp}} = \text{clip}_{[0,1]}\left(\frac{1}{|\Omega|}\sum_{u\in\Omega}\big(2\,p(u) - 1\big)\right). \tag{4}$$

A confident mask whose probabilities sit close to one scores near one; a mask with many pixels near the decision boundary scores lower.

The second cue, S_{stab}, measures consistency under prompt perturbation. The box is jittered K times by a small random shift and scale, the model is run again for each jittered box, and S_{stab} is the mean intersection over union between the original binary mask $\hat{\mathbf{M}}$ and the jittered masks $\hat{\mathbf{M}}_k$:

$$S_{\text{stab}} = \frac{1}{K}\sum_{k=1}^{K}\text{IoU}\Big(\hat{\mathbf{M}}, \hat{\mathbf{M}}_k\Big). \tag{5}$$

A prediction that barely changes when the box moves slightly is treated as more reliable than one that collapses or drifts.

The final score is $Q = \frac{1}{2}\big(S_{\text{sharp}} + S_{\text{stab}}\big)$, with $K = 6$, a shift of up to 0.12 of the box side, and a scale factor within ± 0.10. Q is a heuristic ranking signal, not a calibrated probability: a value of 0.7 does not mean a 70% chance that the mask is correct.

The same score is reused to shortlist candidate regions when no prompt is given. A set of boxes is sampled at varying size and position, each is scored by Q, and the highest-scoring few are returned for the clinician to accept, adjust, or reject. This is a ranking aid for review, not an automatic detector.

TABLE 2. Image-level train / validation / test splits. Each lesion polygon is one promptable sample.

Dataset	Images			Lesion polygons		
	Train	Val	Test	Train	Val	Test
BTXRD	1,493	187	187	1,848	238	232
FracAtlas	573	72	72	730	95	92

IV. EXPERIMENTAL SETUP

A. DATASETS

We evaluate on two public bone radiograph datasets. BTXRD is a primary bone tumor X-ray dataset with polygon annotations for classification, localization, and segmentation [7]. FracAtlas is a fracture radiograph dataset with the same label types [8]; we apply data cleaning and polygon normalization before use. Both are split at the image level, with every polygon from one image kept in the same split (Table 2).

The public metadata were insufficient to verify strict patient-level separation, so images from the same patient could in principle fall into different splits; we treat this as a limitation of the evaluation. For image-level evaluation, if an image contains multiple lesions, the model is run once per lesion box and the probability maps are merged by pixelwise maximum before thresholding at 0.5, and the ground-truth masks are merged by union.

Inputs are resized isotropically and padded to square resolutions of 256×256 or 512×512 to preserve aspect ratio. Images, masks, and prompt maps undergo the same geometric preprocessing.

B. TRAINING DETAILS

All CNN models were implemented in PyTorch and trained on an NVIDIA Tesla T4 GPU with 16 GB memory. PGA-UNet and Attention U-Net were trained separately on each dataset and resolution. In other words, the article studies one shared architecture family instantiated as separate models with separate parameter sets for BTXRD and FracAtlas, rather than a single jointly trained cross-dataset model. The optimizer was AdamW with learning rate 10^{-4} and weight decay 10^{-4}. The loss was the sum of binary cross-entropy and Dice loss. As an exploratory check motivated by the small size of the targets, PGA-UNet was additionally trained at 512×512 with the Dice term replaced by a size-conditioned Tversky loss and, in a separate run, by a Focal Dice loss, with every other setting held fixed; the two replacements are mutually exclusive and are reported later only as a negative control on the training objective. Batch size was 4, the maximum training budget was 150 epochs, and early stopping with patience 15 was used. All main and ablation training runs use the fixed training seed 22120196, which seeds Python, NumPy, and PyTorch randomness on both CPU and CUDA. The primary model-selection criterion was image-level merged validation Dice under the `center_shift` condition.

Training-time augmentation included horizontal flipping and random rotation within $\pm 15°$. The implementation also applied prompt dropout or prompt noise to a subset of training iterations to reduce over-reliance on an ideal prompt. PGA-UNet is trained with `center_mixed`, selecting `center_shift` with probability 0.8 and `center_zoom` with probability 0.2. The prompts are simulated from lesion annotations to provide a controlled prompted-segmentation protocol. The current article should therefore be read as a study under simulated lesion-covering boxes around user-identified lesions, not as an evaluation of unconstrained lesion detection or fully realistic radiologist-drawn boxes. For the fine-tuned SAM-Med2D comparison, the pretrained image encoder was frozen except for its lightweight Adapter layers, while the prompt encoder and mask decoder were fully updated on each dataset under the same split protocol. SAM-Med2D was fine-tuned with box prompts only, using the same center-scaled covering and off-center protocol as PGA-UNet; the point-prompt branch and iterative point refinement were disabled. This box construction differs from the small random-jitter `get_boxes_from_mask` sampling used by the original SAM-Med2D authors, which was not applied here. Validation during SAM-Med2D fine-tuning and the final test split used both prompt scenarios, with batch size 4, a maximum of 150 epochs, early stopping patience 30, and learning rate 10^{-5}. The matched protocol reduces differences caused by prompt distribution and resolution, but does not eliminate differences in model scale, initialization, or optimization. Additional stability experiments were performed using Monte Carlo cross-validation, that is, repeated random image-level splits rather than strict non-overlapping k-fold partitioning. The Monte Carlo experiments keep the training seed fixed at 22120196 and vary only the split seed across 1, 2, 3, and 4.

The training checkpoints also include a small auxiliary quality-regression head that was trained jointly to predict the per-sample Dice. It is detached from the segmentation branch and does not affect the predicted mask. Its output is not used in this article; the self-assessment score of Section III-F is used instead, and Section V reports why.

Two evaluation details are important for interpreting the results. First, prompt generation itself is resolution-agnostic: the plateau heatmap is built and smoothed with a fixed 31×31 Gaussian kernel at the original image resolution, and only the resulting heatmap is resized and padded down to 256×256 or 512×512 together with the image, so the same absolute blur is compressed differently by the two target resolutions rather than being parameterized separately for each. Second, the Monte Carlo cross-validation experiments are auxiliary stability evidence rather than replacements for the fixed train/validation/test split used in the main tables.

C. BASELINES AND METRICS

PGA-UNet is compared with Attention U-Net [4] as the main automatic baseline and with SAM-Med2D [2] as the main prompt-based baseline. Attention U-Net is trained without prompts and is used to characterize how difficult the task is when localization must be solved automatically. SAM-Med2D is evaluated at 256×256 in fine-tuned form with the same dataset splits and the same box prompts as PGA-UNet, making it the primary prompt-matched comparison.

To further isolate whether PGA-UNet's advantage comes from its architecture or simply from having access to the box prompt at all, we add two prompt-matched conventional baselines built on the same Attention U-Net backbone, given the identical box prompt PGA-UNet receives but without PGA-UNet's Gaussian-prior heatmap, Prompt Spatial Gate, or Conditional Attention Decoder. Attention U-Net with a concatenated prompt channel appends the prompt heatmap as a second input channel and predicts on the full image through a plain skip-connection decoder. Attention U-Net on prompt crops instead restricts the network's field of view: the input image is cropped to the prompt box before resizing to the model's input resolution, the plain Attention U-Net predicts a mask on that crop alone, and the prediction is pasted back into the full frame for evaluation. Both baselines are trained under the same covering and off-center box protocol as PGA-UNet, so any remaining gap reflects the architecture rather than an unequal prompt. Cross-model comparisons use the pasted-back full-frame metrics for the prompt-crop baseline, not its crop-frame diagnostic values.

Every comparison table reports Dice and IoU for region overlap, CBL for localization, and HD95 for boundary distance; Dice and IoU are strongly correlated overlap measures and are shown together only so a reader can cross-check. On the small-lesion subsets IoU is low in absolute terms and sensitive to a few boundary pixels, so Dice is the primary overlap metric there. Precision and recall are in the released logs. For a predicted mask centroid (x_p, y_p), a ground-truth centroid (x_g, y_g), and the diagonal length D_{GT} of the ground-truth bounding box, CBL is defined as

$$\mathrm{CBL} = \max\left(0, 1 - \frac{\sqrt{(x_p - x_g)^2 + (y_p - y_g)^2}}{D_{\mathrm{GT}}}\right). \quad (6)$$

HD95 is computed from bidirectional boundary distances. If both masks are empty, HD95 is set to 0. If exactly one mask is empty, HD95 is set to the input side length S (256 or 512 pixels). No samples are discarded. For comparisons that span input resolutions, HD95 is also reported in normalized form, $\mathrm{HD95}/(\sqrt{2}\,S)$, because a raw pixel distance is not comparable across the 128, 256, and 512 frames. Some subgroup comparisons that mix models trained at 256×256 and 512×512 are scored in a shared 512×512 frame: the 256×256 predictions are upsampled before scoring, so every model is measured against identical ground-truth masks rather than in its own native frame.

For the training-free self-assessment score, each lesion prompt produces a score Q and the true thresholded Dice of its own predicted mask; we report the Pearson and Spearman correlation between the two, separately for the covering and off-center conditions. For the candidate-region shortlist, we sample 50 boxes per test image at side length uniform in $[0.15, 0.55]$ of the frame, score each by Q, keep the highest-scoring three and five, and score a retained box by coverage, the fraction of ground-truth lesion pixels it contains; no predicted mask enters this coverage metric.

D. SCOPE OF ANALYSIS

This article focuses on prompt-guided segmentation itself. A lesion-screening classification stage was explored separately and is not part of this article. The self-assessment score and the candidate-region shortlist are reported as auxiliary analyses with a deliberately narrow scope: Q is a heuristic ranking signal rather than a calibrated probability, and the shortlist is a review aid rather than an automatic detector. To keep the article compact, we retain only the subgroup analyses that most directly support the main claim: the role of prompt-guided localization, sensitivity to the controlled prompt displacement defined in this study, and effectiveness on very small lesions. The small-lesion subgroup is defined per dataset as the 50 test images with the smallest total lesion area. For the three-resolution small-lesion comparison, this same set of image stems is held fixed and evaluated at 128, 256, and 512; predictions and ground truth are merged at image level at each resolution, HD95 is reported both in native pixels and in the normalized form above, and degradation is reported relative to the 512 result. Partial-coverage boxes, negative boxes, and unrelated-region boxes are outside the present task scope and are therefore not used to support or reject the paper's claims.

V. RESULTS

Unless stated otherwise, every table reports current-protocol numbers: `center_mixed` training with 80% `center_shift` and 20% `center_zoom`, 150 epochs, model selection by image-level merged validation Dice under `center_shift`, and image-level merged test evaluation. The two test prompt conditions, covering (`center_zoom`) and off-center (`center_shift`), are reported separately whenever a table carries both. The ablation in Section V-G is a single split; a repeated-seed rerun is in progress.

A. MAIN RESULTS ACROSS DATASETS AND RESOLUTIONS

PGA-UNet was trained from scratch at 128×128, 256×256, and 512×512 on each dataset, giving six models. Table 3 reports them under both prompt conditions.

Higher resolution helps at every step. Dice follows $512 > 256 > 128$ on both datasets and under both prompt conditions, and FracAtlas gains more, which fits its thin fracture lesions losing proportionally more detail when the image shrinks. Raw HD95 in pixels cannot be compared across frame sizes, so we also give the normalized form: it stays near 0.03 across the BTXRD resolutions and falls on FracAtlas as the frame grows, so boundary error does not worsen at higher resolution. Each model is evaluated only on its own dataset; no cross-dataset claim is made.

TABLE 3. PGA-UNet across resolutions, image-level merged. Dice, IoU, CBL and HD95 (native px and normalized HD95/($\sqrt{2}\,S$)) given as covering / off-center.

Dataset	Res.	Dice	IoU	$HD95_{px}$	$HD95_n$	CBL
BTXRD	128	0.714 / 0.717	0.572 / 0.577	5.8 / 5.3	0.032 / 0.029	0.859 / 0.857
	256	0.762 / 0.743	0.633 / 0.611	10.9 / 12.8	0.030 / 0.035	0.902 / 0.885
	512	**0.782 / 0.774**	**0.661 / 0.652**	22.8 / 24.5	0.031 / 0.034	**0.918 / 0.912**
FracAtlas	128	0.596 / 0.542	0.445 / 0.389	8.1 / 9.5	0.044 / 0.053	0.849 / 0.790
	256	0.629 / 0.594	0.470 / 0.436	10.6 / 11.7	0.029 / 0.032	0.895 / 0.849
	512	**0.727 / 0.713**	**0.585 / 0.570**	10.5 / 10.4	**0.015 / 0.014**	**0.913 / 0.897**

TABLE 4. Comparison at 512 × 512 under covering prompts. Attention U-Net (no prompt) receives no box; the other rows use the same box PGA-UNet receives. ´Dice is each row minus PGA-UNet.

Dataset	Model	Dice	ΔDice	IoU	HD95	CBL
BTXRD	Attention U-Net (no prompt)	0.517	−0.265	0.418	120.4	0.620
	Attention U-Net + prompt channel	0.760	−0.022	0.635	33.0	0.895
	Attention U-Net + prompt crop	0.741	−0.041	0.615	24.7	0.912
	PGA-UNet	**0.782**	—	**0.661**	**22.8**	**0.918**
FracAtlas	Attention U-Net (no prompt)	0.342	−0.385	0.253	139.9	0.478
	Attention U-Net + prompt channel	0.593	−0.134	0.434	22.9	0.887
	Attention U-Net + prompt crop	0.590	−0.137	0.451	22.0	0.865
	PGA-UNet	**0.727**	—	**0.585**	**10.5**	**0.913**

B. COMPARISON AGAINST AUTOMATIC AND PROMPT-MATCHED ATTENTION U-NET

Left to find the lesion on its own, Attention U-Net manages a Dice of only 0.517 on BTXRD and 0.342 on FracAtlas at 512×512, with HD95 past 120 pixels on both (Table 4). That number is not an architecture verdict, since this model never sees the box; it measures how hard automatic localization is, which is exactly the step the prompt removes.

Give the same box to a plain Attention U-Net and the picture changes but a gap remains (Table 4, ΔDice column). PGA-UNet leads both prompt-matched variants on both datasets, by a small margin on BTXRD and a much wider one on FracAtlas, with lower HD95 than either.

So localization is a large part of the difficulty here, but having the box is not the whole story. The way PGA-UNet consumes that box buys a further, consistent margin over a conventional network given the identical input.

C. ATTENTION U-NET TOP-DICE AND BOTTOM-DICE SUBSETS

Where does that margin come from? We ranked the test images by Attention U-Net Dice, kept the top 50 and the bottom 50 per dataset, and re-ran all four models on those exact stems (Table 5, covering prompts). On the top-50 the prompt-guided models bunch together. On the bottom-50 Attention U-Net collapses, to 0.031 on BTXRD and 0.169 on FracAtlas, while PGA-UNet still holds 0.712 and 0.687. The two conventional prompt baselines recover as well, but less: on the bottom-50 PGA-UNet leads them by 0.04 to 0.07 Dice on BTXRD and by 0.10 to 0.19 on FracAtlas. So a prompt is necessary but not sufficient; the benefit of prompt-conditioned processing is concentrated exactly where automatic localization breaks

TABLE 5. Top-Dice and bottom-Dice subsets at 512 × 512, 50 images per group, covering prompts. Groups are defined by plain Attention U-Net.

Dataset	Group	Model	Dice	IoU	HD95	CBL
BTXRD	Top-50	Attention U-Net	0.872	0.775	35.0	0.945
		AttUNet + channel	0.846	0.740	24.3	0.942
		AttUNet + crop	0.844	0.739	27.5	0.946
		PGA-UNet	0.862	0.763	19.2	0.944
	Bottom-50	Attention U-Net	0.031	0.016	277.8	0.105
		AttUNet + channel	0.668	0.541	43.3	0.831
		AttUNet + crop	0.652	0.518	17.6	0.875
		PGA-UNet	**0.712**	**0.589**	23.2	**0.879**
FracAtlas	Top-50	Attention U-Net	0.493	0.365	85.7	0.640
		AttUNet + channel	0.605	0.447	26.6	0.887
		AttUNet + crop	0.658	0.515	24.9	0.881
		PGA-UNet	**0.754**	**0.615**	10.6	**0.927**
	Bottom-50	Attention U-Net	0.169	0.106	182.3	0.315
		AttUNet + channel	0.574	0.416	25.3	0.875
		AttUNet + crop	0.509	0.369	26.0	0.834
		PGA-UNet	**0.687**	**0.535**	11.8	**0.896**

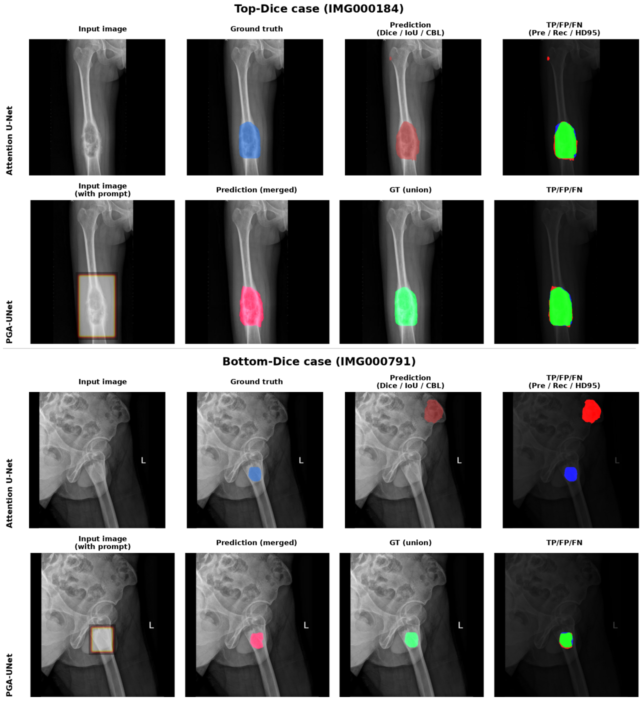

FIGURE 3. Qualitative comparison between Attention U-Net and PGA-UNet on BTXRD for one Attention U-Net top-Dice case (IMG000184) and one bottom-Dice case (IMG000791). The rows use identical image stems; for PGA-UNet the input-image and box-prompt panels are merged into one.

down. These groups are defined by Attention U-Net, not by any objective notion of difficulty.

D. MATCHED COMPARISON AGAINST SAM-MED2D

Both models get the same box at 256 × 256 (Table 6). Zero-shot, SAM-Med2D barely works on bone radiographs: 0.255 Dice on BTXRD, 0.262 on FracAtlas. Fine-tuning on the same split and prompt protocol lifts it to 0.630 and 0.563, but PGA-UNet at that resolution is at 0.762 and 0.629 under covering prompts, ahead by 0.132 and 0.066 Dice and by a similar amount in CBL. SAM-Med2D pulls ahead in only one place, boundary distance on FracAtlas, where its fine-tuned

TABLE 6. Comparison with SAM-Med2D at 256 × 256, image-level merged, covering prompts.

Dataset	Model	Dice	IoU	HD95	CBL
BTXRD	SAM-Med2D zero-shot	0.255	0.156	41.2	0.734
	SAM-Med2D fine-tuned	0.630	0.497	17.3	0.836
	PGA-UNet	**0.762**	**0.633**	**10.9**	**0.902**
FracAtlas	SAM-Med2D zero-shot	0.262	0.156	21.3	0.752
	SAM-Med2D fine-tuned	0.563	0.417	**8.0**	0.846
	PGA-UNet	**0.629**	**0.470**	10.6	**0.895**

TABLE 7. Small-lesion subset, PGA-UNet vs SAM-Med2D at the matched 256 × 256 resolution. 50 images per dataset, covering prompts, scored in a shared 512 × 512 frame.

Dataset	Model	Dice	IoU	HD95	CBL
BTXRD	SAM-Med2D zero-shot 256	0.139	0.082	62.4	0.515
	SAM-Med2D fine-tuned 256	0.256	0.170	123.8	0.561
	PGA-UNet 256	**0.705**	**0.559**	**14.7**	**0.877**
FracAtlas	SAM-Med2D zero-shot 256	0.205	0.123	33.5	0.659
	SAM-Med2D fine-tuned 256	0.371	0.256	62.1	0.736
	PGA-UNet 256	**0.648**	**0.491**	**20.6**	**0.897**

HD95 of 8.0 pixels beats PGA-UNet's 10.6. The covering-to-off-center change is a separate question; Section V-F takes it up.

E. PERFORMANCE ON SMALL LESIONS

If keeping weak evidence alive through the network matters anywhere, it is here. The small-lesion subset is the 50 test images per dataset with the least total lesion area, 7 to 433 pixels on BTXRD and 145 to 944 on FracAtlas. All scores are computed in a shared 512 × 512 frame.

a: Against SAM-Med2D at the matched resolution

SAM-Med2D struggles badly at this size (Table 7). Fine-tuned at 256, it reaches only 0.256 Dice on BTXRD and 0.371 on FracAtlas, and zero-shot it is close to nothing, while PGA-UNet at the same 256 holds 0.705 and 0.648 with far lower HD95.

b: Against prompt-matched Attention U-Net at 512

Given the same box at 512, the two conventional prompt baselines trail PGA-UNet by 0.04 on BTXRD and by 0.11 to 0.16 on FracAtlas, and the prompt-free Attention U-Net drops to around 0.28 (Table 8). The prompt-crop variant reaches a lower HD95 on BTXRD, where its tight crop suits a compact target, but not on FracAtlas.

c: Resolution on the small-lesion stems

Resolution bites hardest here. On the fixed small-lesion stems, PGA-UNet Dice falls from 0.766 at 512 to 0.631 at 128 on BTXRD and from 0.720 to 0.609 on FracAtlas, a 512-to-128 loss of 0.135 and 0.111, more than on the full test set (Table 9).

Read narrowly, the smallest lesions are the stress case for the whole approach, and both the prompt conditioning and the input resolution count for more here than on the full set.

TABLE 8. Small-lesion subset at 512 × 512, PGA-UNet vs Attention U-Net baselines. 50 images per dataset, covering prompts.

Dataset	Model	Dice	IoU	HD95	CBL
BTXRD	Attention U-Net (no prompt)	0.283	0.219	178.2	0.283
	Attention U-Net + prompt channel	0.728	0.591	14.2	0.879
	Attention U-Net + prompt crop	0.724	0.602	**5.5**	**0.909**
	PGA-UNet	**0.766**	**0.637**	14.0	0.897
FracAtlas	Attention U-Net (no prompt)	0.291	0.216	139.4	0.401
	Attention U-Net + prompt channel	0.610	0.449	21.9	0.888
	Attention U-Net + prompt crop	0.564	0.425	22.6	0.855
	PGA-UNet	**0.720**	**0.578**	**8.3**	**0.907**

TABLE 9. PGA-UNet on the small-lesion subset across resolutions, 50 fixed stems per dataset. Dice, CBL and HD95 (native px / normalized) as covering / off-center. ´ is the covering Dice loss relative to 512.

Dataset	Res.	Dice	IoU	CBL	$HD95_{px}$ / $_n$	ΔDice
BTXRD	128	0.631 / 0.662	0.488 / 0.517	0.738 / 0.758	3.9 / 0.022	−0.135
	256	0.714 / 0.695	0.570 / 0.548	0.869 / 0.849	7.2 / 0.020	−0.052
	512	**0.766 / 0.757**	**0.637 / 0.627**	**0.897 / 0.890**	14.0 / 0.019	—
FracAtlas	128	0.609 / 0.543	0.462 / 0.393	0.828 / 0.757	8.9 / 0.049	−0.111
	256	0.642 / 0.610	0.484 / 0.451	0.887 / 0.845	10.3 / 0.029	−0.079
	512	**0.720 / 0.712**	**0.578 / 0.570**	**0.907 / 0.893**	8.3 / 0.012	—

F. SENSITIVITY TO PROMPT DISPLACEMENT

Table 10 moves the box off-center under one fixed rule. PGA-UNet loses 0.008 Dice on BTXRD and 0.014 on FracAtlas, with CBL down by 0.006 and 0.015; on the small-lesion subset the loss stays under 0.01 on both. The conventional prompt baselines lose a comparable or slightly larger amount, and fine-tuned SAM-Med2D about 0.03. This is sensitivity to one displacement rule fixed in advance. It says nothing about arbitrary radiologist-drawn boxes, and partial or off-target boxes were never tested.

G. ABLATION STUDY

The ablation drops one factor at a time from the full model, Gaussian prompt with PSG and CAD, and separately swaps the Gaussian prompt for a binary box (Table 11). The full model has the best Dice in three of the four dataset and prompt cells. The exception is BTXRD under covering prompts, where the binary-box variant is 0.006 Dice higher, within single-split noise, while the full model still keeps the lowest HD95 and the highest CBL there. The Gaussian prompt's advantage is clearest under off-center prompts, where the full model wins on both datasets, and on FracAtlas, where it wins every configuration on every metric. The aim is to test the combination as a system, not to claim a unique internal mechanism. These are single-split numbers; a repeated-seed rerun is in progress to confirm the ordering.

H. MONTE CARLO CROSS-VALIDATION

One fixed split could flatter or hurt the method by chance, so PGA-UNet at 512 × 512 was retrained on four fresh random image-level splits, with the training seed held at 22120196 and only the split seed changed (Table 12). Dice varies by at most 0.011 across splits on either dataset and either prompt condition; CBL by at most 0.013. The FracAtlas repeated-split mean, 0.733, sits just above the fixed-split test result of 0.727, so if anything the reported split is slightly harder than typical. This is evidence that PGA-UNet is stable across splits, not that it beats any baseline.

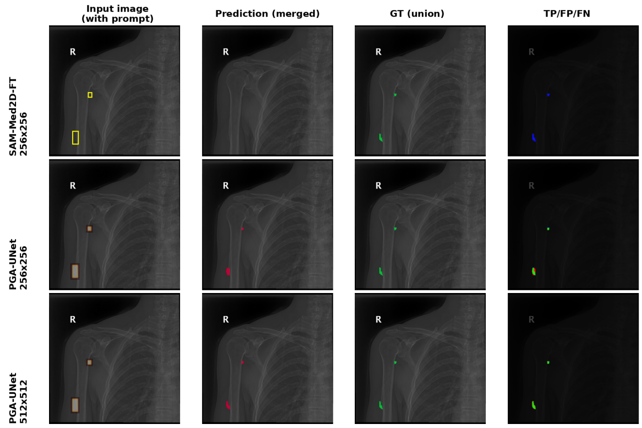

FIGURE 4. Representative small-lesion comparison on BTXRD (IMG000868, two small polygons): fine-tuned SAM-Med2D at 256 × 256, PGA-UNet at 256 × 256, and PGA-UNet at 512 × 512. The input-image and box-prompt panels are merged into one.

TABLE 10. Sensitivity to prompt displacement. Every model under covering (z) and off-center (s) prompts. Attention U-Net (no prompt) is prompt-invariant. AttUNet rows and PGA-UNet at 512 × 512; SAM-Med2D fine-tuned at 256 × 256, image-level merged.

Dataset	Model	Dice z/s	ΔDice	IoU z/s	HD95 z/s	CBL z/s
BTXRD	Attention U-Net (no prompt)	0.517 / 0.517	0.000	0.418 / 0.418	120.4 / 120.4	0.620 / 0.620
	Attention U-Net + prompt channel	0.760 / 0.748	−0.012	0.635 / 0.626	33.0 / 32.6	0.895 / 0.890
	Attention U-Net + prompt crop	0.741 / 0.733	−0.008	0.615 / 0.607	24.7 / 26.4	0.912 / 0.896
	SAM-Med2D fine-tuned	0.630 / 0.601	−0.029	0.497 / 0.471	17.3 / 17.5	0.836 / 0.813
	PGA-UNet	**0.782 / 0.774**	**−0.008**	**0.661 / 0.652**	22.8 / 24.5	**0.918 / 0.912**
FracAtlas	Attention U-Net (no prompt)	0.342 / 0.342	0.000	0.253 / 0.253	139.9 / 139.9	0.478 / 0.478
	Attention U-Net + prompt channel	0.593 / 0.588	**−0.005**	0.434 / 0.428	22.9 / 24.0	0.887 / 0.860
	Attention U-Net + prompt crop	0.590 / 0.563	−0.027	0.451 / 0.428	22.0 / 23.8	0.865 / 0.833
	SAM-Med2D fine-tuned	0.563 / 0.532	−0.031	0.417 / 0.389	8.0 / 8.4	0.846 / 0.816
	PGA-UNet	**0.727 / 0.713**	−0.014	**0.585 / 0.570**	**10.5 / 10.4**	**0.913 / 0.897**

I. EFFICIENCY

PGA-UNet is small, and all 2.955M of its parameters are trainable (Table 13). At the matched 256 × 256 resolution it is about 92 times lighter than fine-tuned SAM-Med2D in parameters, 12 times lighter in FLOPs, 215 times smaller on disk, and 6 times faster on GPU. These are measured costs on one implementation, one GPU, and two resolutions, nothing more.

J. EXPLORATORY COMPARISON OF SEGMENTATION LOSSES

Because the targets are small, we also tried swapping the Dice term for a size-conditioned Tversky loss, and separately for a Focal Dice loss, everything else held fixed (Table 14). Neither helped. The default Dice + BCE is best or tied on both datasets and both prompt conditions, and Focal Dice is a plain regression on FracAtlas, with HD95 above 100 pixels. We keep this as a negative control: under the current protocol the default objective is already a good choice, and neither alternative is adopted.

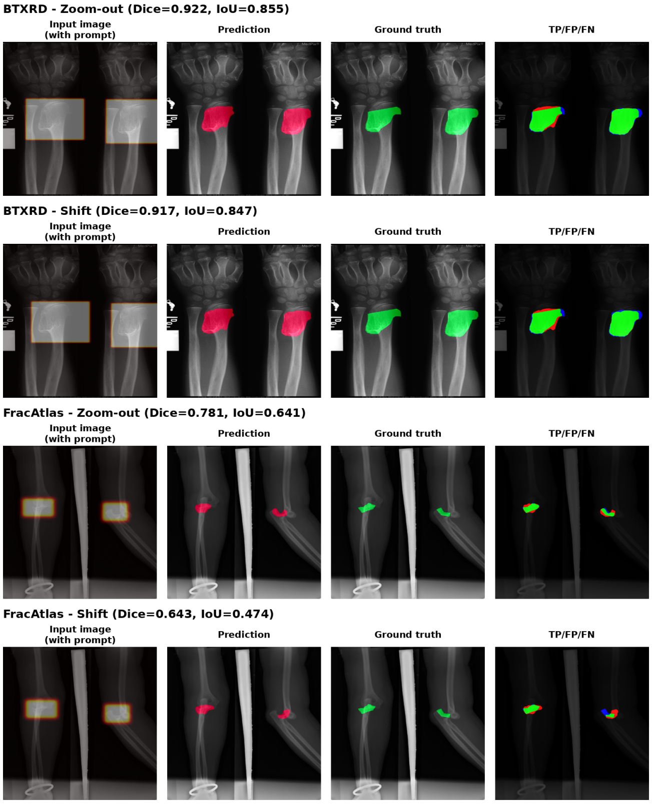

FIGURE 5. Representative behavior under prompt perturbation on both datasets. Thin fracture patterns remain harder than compact lesions even when the box still covers the target.

TABLE 11. Ablation at 512 × 512, image-level merged, single split. Dice, IoU, HD95 and CBL as covering / off-center. A repeated-seed rerun is in progress.

Dataset	Configuration	Dice	IoU	HD95	CBL
BTXRD	CAD only	0.771 / 0.759	0.649 / 0.635	23.1 / 24.2	0.915 / 0.905
	PSG only	0.779 / 0.771	0.660 / 0.651	29.4 / 28.4	0.908 / 0.902
	PSG + vanilla attention gate	0.769 / 0.757	0.645 / 0.633	26.4 / 28.7	0.906 / 0.896
	Full model + binary box prompt	**0.788** / 0.770	**0.668** / 0.651	23.1 / 28.5	0.914 / 0.900
	Full PGA-UNet (Gaussian + PSG + CAD)	0.782 / **0.774**	0.661 / **0.652**	**22.8** / **24.5**	**0.918** / **0.912**
FracAtlas	CAD only	0.696 / 0.685	0.551 / 0.540	18.0 / **12.0**	0.903 / 0.885
	PSG only	0.635 / 0.611	0.479 / 0.459	14.5 / 14.8	0.894 / 0.857
	PSG + vanilla attention gate	0.690 / 0.691	0.548 / 0.547	14.0 / 14.1	0.892 / 0.881
	Full model + binary box prompt	0.668 / 0.657	0.514 / 0.505	12.0 / 13.2	0.904 / 0.886
	Full PGA-UNet (Gaussian + PSG + CAD)	**0.727** / **0.713**	**0.585** / **0.570**	**10.5** / 10.4	**0.913** / **0.897**

K. FAILURE MODES

PGA-UNet still fails, and the failures have a pattern. In the qualitative review, low Dice clusters on lesions that are elongated or branching, that sit in cluttered anatomy such as the shoulder or clavicle, or that barely stand out from the bone around them. A covering box does not fix a case where the local image evidence is itself poor.

L. TRAINING-FREE SELF-ASSESSMENT AND CANDIDATE SUGGESTION

Without any ground truth, the score Q from Section III-F still tracks the true per-prompt Dice (Table 15). Its Spearman correlation with the true Dice is 0.70 covering and 0.60 off-center on BTXRD, and 0.65 and 0.48 on FracAtlas. The two cues pull their weight: mask decisiveness alone reaches a Spearman near 0.4, jitter stability near 0.5 to 0.6, and their mean is at least as good as either. We also trained a small head to regress the Dice directly; it collapsed to a near-constant output, Spearman under 0.2 on both datasets, and is not used.

Turned around, Q can rank 50 random candidate boxes and keep the top few (Table 16, Figure 8). At least one box in the top five covers more than half the lesion for 67% of BTXRD images and 82% of FracAtlas images, and the best of the five covers 0.84 and 0.90 of the lesion area on average. This is a review aid and nothing stronger: Q is a heuristic, not a calibrated probability, and the shortlist is not a detector.

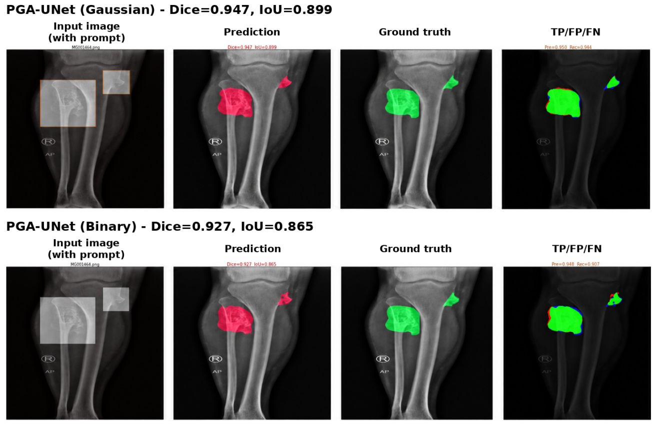

FIGURE 6. Representative comparison between the Gaussian plateau prompt and a binary box prompt within the full architecture.

TABLE 12. Monte Carlo cross-validation of PGA-UNet at 512 × 512, four repeated random image-level splits, mean ± standard deviation.

Dataset	Prompt	Dice	IoU	HD95	CBL
BTXRD	Covering	0.769 ± 0.005	0.647 ± 0.006	25.0 ± 2.5	0.905 ± 0.005
	Off-center	0.762 ± 0.004	0.639 ± 0.007	25.7 ± 2.0	0.901 ± 0.007
FracAtlas	Covering	0.733 ± 0.007	0.595 ± 0.007	14.1 ± 2.7	0.909 ± 0.006
	Off-center	0.728 ± 0.011	0.590 ± 0.010	15.6 ± 3.9	0.904 ± 0.013

VI. DISCUSSION

The question this article set out to answer was never whether a prompt helps in the abstract. It was whether the way a model handles the prompt still matters once a coarse box has already narrowed the search. Attention U-Net without a box answers a different question, and its low Dice, near 0.34 on FracAtlas, measures how much of the task is just finding the lesion. The comparisons that actually test the design give the same box to a plain Attention U-Net and to fine-tuned SAM-Med2D. In both, PGA-UNet keeps a margin: modest on BTXRD, clearly larger on FracAtlas, and largest of all on the Attention U-Net bottom-Dice group and on the smallest lesions. That is the pattern one would expect if PSG and CAD are doing their intended job, holding the prompt active through the encoder and decoder so a faint signal is not lost to downsampling. The single-split per-component ablation (Table 11) is consistent with this, with the Gaussian prompt ahead under off-center prompts and on FracAtlas; a repeated-seed rerun to confirm the ordering is in progress.

Two side results are worth stating without hedging. The

TABLE 13. Efficiency in the tested environment (NVIDIA Tesla T4, batch size 1).

Model	Params	GFLOPs	Size	GPU ms	CPU ms
PGA-UNet 256	2.96M	7.74	11.4 MB	8.5	112
PGA-UNet 512	2.96M	30.97	11.4 MB	18.5	375
SAM-Med2D 256	271.24M	92.02	2443 MB	50.1	1324

TABLE 14. Exploratory segmentation-loss comparison for PGA-UNet at 512 × 512, everything else fixed. Dice, IoU, CBL as covering / off-center. The default is not changed.

Dataset	Loss	Dice	IoU	HD95	CBL
BTXRD	Dice + BCE (default)	**0.782 / 0.774**	**0.661 / 0.652**	22.8 / 24.5	**0.918 / 0.912**
	Size-conditioned Tversky	0.778 / 0.767	0.660 / 0.649	25.2 / 30.3	0.916 / 0.903
	Focal Dice	0.773 / 0.765	0.649 / 0.642	25.6 / 25.6	0.913 / 0.907
FracAtlas	Dice + BCE (default)	**0.727 / 0.713**	**0.585 / 0.570**	**10.5 / 10.4**	0.913 / **0.897**
	Size-conditioned Tversky	0.709 / 0.680	0.561 / 0.530	11.0 / 12.2	**0.918** / 0.888
	Focal Dice	0.642 / 0.578	0.493 / 0.425	105.5 / 110.1	0.802 / 0.732

loss comparison is a clean negative: a size-conditioned Tversky loss or a Focal Dice loss did not help, and Focal Dice on FracAtlas made things worse, so the plain Dice + BCE objective stays. The self-assessment score is a modest positive: Q correlates with the true Dice at a rank level around 0.5 to 0.7 without any label, and it does so where a small learned regression head does not, since that head simply predicted the mean. Q is still a heuristic. It is not calibrated, and the candidate-box shortlist built from it is a way to focus a clinician's attention, not a detector.

The limits of the study are concrete. Inputs are resized to a fixed square, and the resolution results show that this costs real accuracy on the smallest lesions. The prompts are drawn from annotations, not by radiologists, and the protocol assumes the user has already found the region and drawn a box that covers it; partial, negative, and off-target boxes were never in scope. BTXRD and FracAtlas are trained apart, with no joint model and no cross-dataset test, and patient-level separation could not be confirmed from the public metadata. The Monte Carlo runs show stability across splits, not superiority. The ablation is mid-transition from one split to many. This round also leaves SAM-Med2D as the sole promptable foundation reference; a comparison against more recent promptable medical models is left for later. And the gains under prompt perturbation, though consistent with the design intent, are not backed by feature-level probing, so the ablation, once finished, will speak to the usefulness of the design as a whole rather than to the exact mechanism inside it.

VII. CONCLUSION

This article presented PGA-UNet, a lightweight prompt-guided CNN for interactive bone X-ray lesion segmentation. The design turns a coarse lesion-covering box into a Gaussian plateau prior and keeps it active through an encoder-side Prompt Spatial Gate and a decoder-side Conditional Attention Decoder. On BTXRD and FracAtlas, evaluated under controlled covering and off-center prompts, PGA-UNet is consistently ahead of prompt-matched conventional baselines

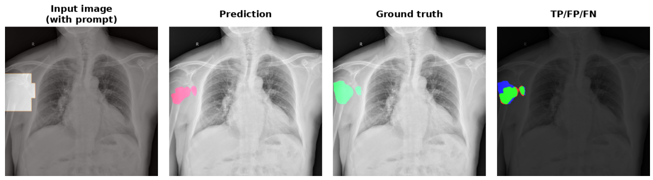

FIGURE 7. Representative failure in an anatomically overlapping region. Even with a covering prompt the boundary remains hard because the local structures are visually confounded.

TABLE 15. Spearman correlation between each training-free cue and the true per-prompt Dice. $Q = \text{mean}(S_{\text{sharp}}, S_{\text{stab}})$.

	BTXRD		FracAtlas	
Cue	cover	off-center	cover	off-center
$S_{\mathtt{sharp}}$ (mask decisiveness)	0.42	0.39	0.47	0.43
$S_{\mathtt{stab}}$ (jitter stability)	0.64	0.56	0.63	0.45
Q	**0.70**	**0.60**	**0.65**	**0.48**

and of fine-tuned SAM-Med2D given the same box, with the margin concentrated where automatic localization fails and where lesions are smallest, and it does so with about 92 times fewer parameters than SAM-Med2D. Repeated-split experiments show the trend is stable, and a training-free self-assessment score gives a usable, if uncalibrated, no-ground-truth ranking of prediction quality.

The conclusions are bounded by the protocol. They concern simulated lesion-covering boxes derived from annotations, not unconstrained detection, partial or off-target boxes, radiologist-drawn prompts, calibrated confidence, patient-level generalization, or cross-dataset transfer. Future work should complete the repeated-seed ablation, add a broader comparison with recent promptable medical models, replace the simulated boxes with looser and more realistic prompt protocols, and turn the self-assessment score into a calibrated signal that could then drive candidate-region suggestion in a clinical review loop.

DATA AVAILABILITY

BTXRD and FracAtlas are public datasets cited in this article. The manuscript reports experiments on image-level train/validation/test splits derived from those datasets. Code and split details should be released with the final reproducibility package.

AUTHOR CONTRIBUTIONS

Thong Luc: conceptualization, methodology, software, investigation, formal analysis, visualization, writing of the original draft. Nguyen Huu Binh: software support, data preparation, experiment verification, review and editing. Ly Quoc Ngoc: supervision, validation, review and editing.

ETHICS STATEMENT

This study used public de-identified radiograph datasets and did not involve prospective data collection, patient contact, or intervention.

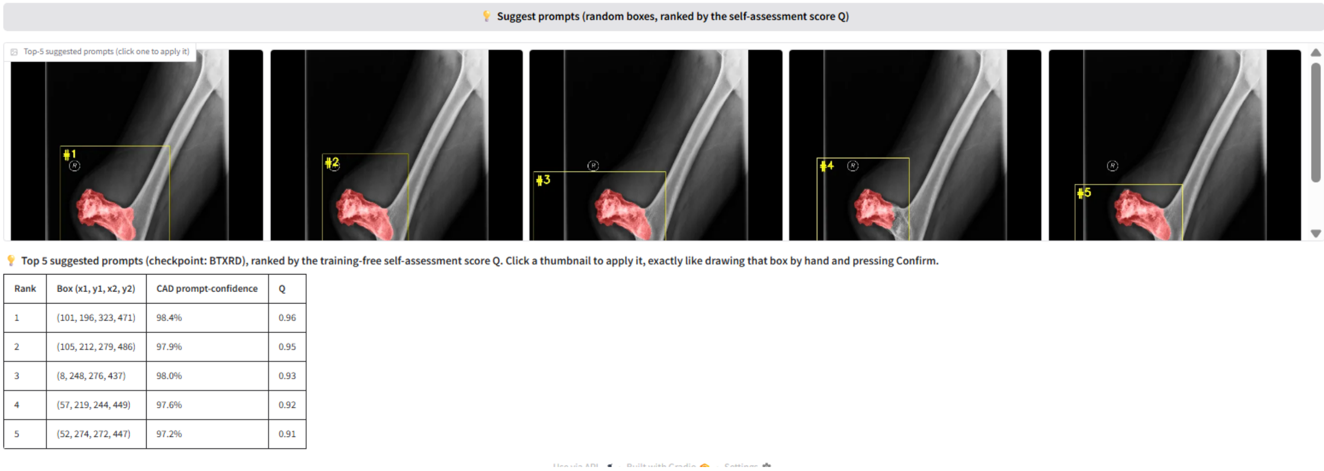

FIGURE 8. **Candidate-region shortlist on a BTXRD case. Fifty boxes are sampled with no prompt, scored by the self-assessment score *Q*, and the five highest-scoring boxes are shown for the clinician to accept, adjust, or reject. A FracAtlas case is shown in the supplementary material.**

TABLE 16. **Candidate-region shortlist ranked by *Q*. Fifty prompt-free boxes per image are scored and the top *k* kept. Coverage is the fraction of lesion pixels inside a box; full-coverage counts images whose best shortlisted box covers more than half the lesion.**

Dataset	List	Mean cov.	Best-of-list cov.	Full-cov. rate
BTXRD	top-3	0.55	0.77	57.8%
	top-5	0.52	0.84	66.8%
FracAtlas	top-3	0.68	0.87	77.8%
	top-5	0.62	0.90	81.9%

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REFERENCES

[1] A. Kirillov *et al.*, "Segment Anything," in *Proc. IEEE/CVF Int. Conf. Comput. Vis.*, 2023, pp. 4015–4026.
[2] J. Cheng *et al.*, "SAM-Med2D," *arXiv preprint arXiv:2308.16184*, 2023.
[3] O. Ronneberger, P. Fischer, and T. Brox, "U-Net: Convolutional Networks for Biomedical Image Segmentation," in *MICCAI*, 2015, pp. 234–241.
[4] O. Oktay *et al.*, "Attention U-Net: Learning Where to Look for the Pancreas," in *Medical Imaging with Deep Learning*, 2018.
[5] N. Xu, B. Price, S. Cohen, J. Yang, and T. S. Huang, "Deep Interactive Object Selection," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, 2016, pp. 373–381.
[6] G. Wang *et al.*, "DeepIGeoS: A Deep Interactive Geodesic Framework for Medical Image Segmentation," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 41, no. 7, pp. 1559–1572, 2019.
[7] S. Yao *et al.*, "A Radiograph Dataset for the Classification, Localization, and Segmentation of Primary Bone Tumors," *Scientific Data*, vol. 12, no. 1, p. 88, 2025.
[8] I. Abedeen *et al.*, "FracAtlas: A Dataset for Fracture Classification, Localization and Segmentation of Musculoskeletal Radiographs," *Scientific Data*, vol. 10, no. 1, p. 521, 2023.
[9] J. Ma *et al.*, "Segment Anything in Medical Images," *Nat. Commun.*, vol. 15, no. 1, p. 654, 2024.
[10] F. Isensee, P. F. Jaeger, S. A. A. Kohl, J. Petersen, and K. H. Maier-Hein, "nnU-Net: a self-configuring method for deep learning-based biomedical image segmentation," *Nat. Methods*, vol. 18, pp. 203–211, 2021.
[11] H. E. Wong, M. Rakic, J. Guttag, and A. V. Dalca, "ScribblePrompt: Fast and Flexible Interactive Segmentation for Any Biomedical Image," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2024, pp. 207–229.
[12] G. Dong *et al.*, "An efficient segment anything model for the segmentation of medical images," *Sci. Rep.*, vol. 14, art. 19425, 2024.
[13] J. Wu *et al.*, "Medical SAM adapter: Adapting segment anything model for medical image segmentation," *Med. Image Anal.*, vol. 102, art. 103547, 2025.

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